LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Thermocouple principle, thermocouple materials and combinations
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Thermocouple principle, thermocouple materials and combinations. Types of thermocouples, Thermocouple harness assemblies. Cold junction temperature compensation
5. Specific Objectives	15.1.5.3 Trainee will be able to explain thermocouple principle and use to measure EGT of engines.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of thermocouple harness assembly connection and cold junction compensation verification; Safety: Ensure engine is cold before handling probes; Instructor supervision: Monitor probe insertion depth and electrical continuity checks.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a thermocouple in an aircraft engine?
(b) Name one material commonly used in thermocouple wires.
(c) How is engine temperature typically measured in modern aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Thermocouple principle of operation • Seebeck effect explanation • Voltage generation mechanism • Application for EGT measurement	L & Ds	PPT, IB	15 min
(b)	• Thermocouple materials and combinations • Type K and Type J characteristics • Thermocouple harness assemblies • Wire insulation and shielding types	L & Ds	PPT, PDF	20 min
(c)	• Cold junction temperature compensation • Reference junction requirements • Compensation methods in EGT probes • Impact of ambient temperature errors	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the Seebeck effect generates a measurable voltage in a thermocouple.
(b) Describe the role of cold junction compensation in accurate EGT measurement.
(c) Identify two critical factors affecting thermocouple harness assembly integrity.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between Type K and Type J thermocouples and their specific applications in aircraft engine monitoring, referencing the next lesson on rotational speed measurement principles.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principle of operation and constructional details of rotational speed measurement. Principle of operation and constructional details. Practical on rotational speed measurement.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min